IGCSE 0607 Practice Questions — E5.5

Angles

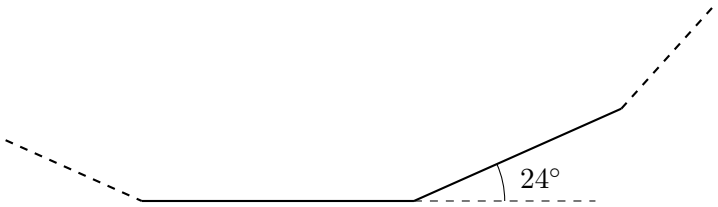
Total Questions: 6 — **Total Marks:** 34 Marks

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1 The diagram shows part of a regular polygon with n sides.
The exterior angle of this polygon is 24°.



(a) Calculate the value of n.

$n =$ ..... **[2]**

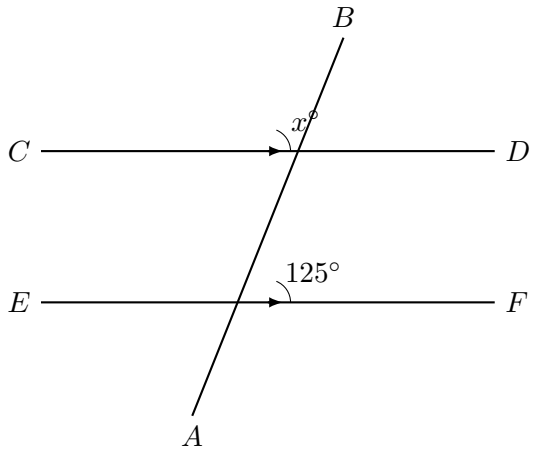
(b) Calculate the sum of the interior angles of this polygon.

..... ° **[2]**

(c) A different regular polygon has an interior angle of 165°.
Work out the number of sides of this polygon.

..... **[2]**

2 The diagram shows a straight line AB intersecting two parallel lines, CD and EF.



(a) Find the value of x.

$x =$ ..... **[1]**

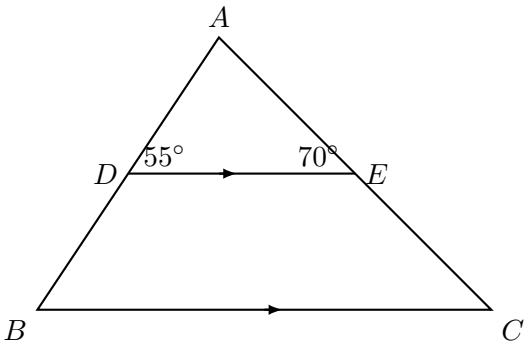
(b) Give a geometric reason for your answer to part (a).

..... **[1]**

(c) Calculate the interior angle of a regular decagon (10-sided polygon).

..... ° **[1]**

3 The diagram shows a triangle ABC with a line DE drawn parallel to BC.
D lies on AB and E lies on AC.
Angle $ADE = 55°$ and angle $AED = 70°$.



(a) Calculate angle DAE.

Angle $DAE =$ ..... ° **[2]**

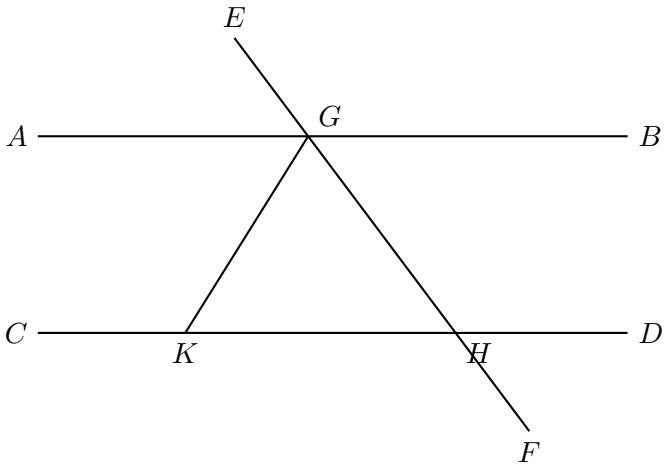
(b) Calculate angle ABC.

Angle $ABC =$ ..... ° **[1]**

(c) Explain why triangle ADE is an isosceles triangle.

[2]

4 In the diagram, the line AB is parallel to the line CD.
The transversal EF intersects AB at G and CD at H.
The point K lies on CD.



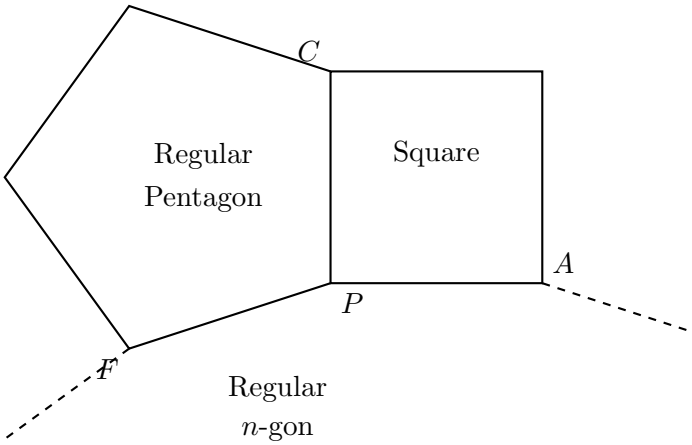
(a) Angle $EGB = (5x + 10)°$ and angle $GHD = (7x - 30)°$.
Find the value of x.
Give a geometrical reason for your working.

$x =$ ..... , Reason: ..... **[3]**

(b) Angle $HGK = 35°$.
Calculate angle GKH.

..... ° **[3]**

5 The diagram shows three regular polygons meeting at the point P.
The polygons are a square, a regular pentagon, and a regular polygon with n sides.
The square shares the side PC with the pentagon and the side PA with the n-gon.
The pentagon shares the side PF with the n-gon.
They meet with no gaps and do not overlap.



(a) Calculate the interior angle of a regular pentagon.

..... ° **[2]**

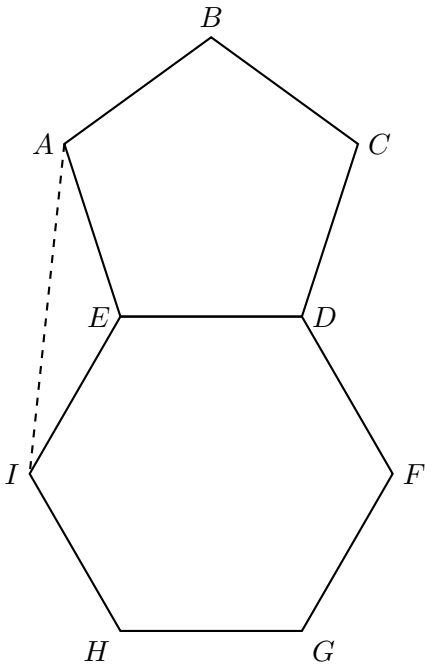
(b) Find the value of n.

$n =$ ..... **[3]**

(c) The vertices A and F of the n-gon are joined by a straight line.
Calculate angle PFA.

..... ° **[2]**

6 The diagram shows a regular pentagon $ABCDE$ and a regular hexagon $EDFGHI$ sharing a common edge ED.



(a) Calculate angle CDE.

Angle $CDE =$ ..... ° **[2]**

(b) Calculate angle AEI.

Angle $AEI =$ ..... ° **[2]**

(c) Calculate angle EAI.

Angle $EAI =$ ..... ° **[3]**